

Hybrid Renewable-based Reliable and Cost-effective Rural Electrification in Southern Africa

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Abstract— The national electrification rate of South Africa is about 75%, however only 55% of rural areas have access to electricity compared to 88% in urban areas [1]. Off grid renewable energy systems (HRES) are becoming promising alternatives to those regions and can replace conventional power systems such as diesel generators or grid power. They incorporate two or more renewable energy systems which complement each other making the system more efficient and reliable. In addition to the HRES, a battery storage system and a diesel generator (DG) can be added for uninterrupted power supply [2][3] when neither renewable energy power sources can meet the load demand. However, the components of the system must be sized optimally to provide a cost effective and reliable solution. Therefore, a techno-economic analysis comprising of various hybrid configurations is required. In this paper, eleven HRES systems comprising of solar PV, wind turbines, diesel generator and battery storage were designed using HOMER for the electrification of two rural villages in Sub Saharan Africa namely Ncera and Boipuso. Optimization was carried out using HOMER and based on the results, a Solar/Wind/Diesel Generator/ Battery system was selected as the most cost-effective solution for both villages. The optimal system for Ncera consists of a 7.80 kW rated PV system, a 20-kW rated wind turbine, a 50-kW convertor, a 44-kW diesel generator with 45 units of Trojan IDN29 batteries capable of supplying electricity with a LCOE of \$0.173/kWh. The optimal system capacities for Boipuso include a 70.3 kW rated PV system, two 20kW rated wind turbines, a 100-kW rated diesel generator 102 strings of batteries with a 75kW rated converter and can supply electricity at a LCOE of \$0.210/kWh.

Keywords—Hybrid renewable energy system, solar, wind, diesel, battery, NPC, COE, sensitivity analysis

I. INTRODUCTION

Energy is the key input in economic growth and it is widely recognized that access to modern energy plays an important role in poverty alleviation and sustainable development. With increased technological advancement in the field and decrease in the LCOE of renewable energy systems [4], the utilization of renewable energy resources provides a promising alternative for rural electrification in South Africa. The important aspect of such systems is that they must be economically feasible while being reliable in meeting the load requirement and this is directly related to the sizing of the system components.

Various optimization techniques have been used throughout the years which include classical techniques such as probabilistic [5] and iterative methods [6] to modern techniques involving Particle swarm optimization [7], Generic Algorithms [8] and software simulation tools such as HOMER [9] [10] [11] [12] and Hybrid2 etc.

In this study, the electrification of two rural villages using an HRES system were considered in HOMER. The villages were selected from two regions namely, the Eastern Cape province of South Africa, which has the lowest grid connection rates at 64.5% when compared to the other provinces [13] and Lesotho, where only 6% of rural access to electricity is accounted for [14]. Both regions have abundant solar and wind resources which provide a high technical potential for the implementation of a Solar PV and wind turbine HRES [1].

II. SELECTION AND RESOURCE ASSESSMENT OF VILLAGES

The first site is the unelectrified village of Ncera [15] located in the coastal region of the Eastern Cape. The second site is the Boipuso Village located in the Mohale's Hoek District of Lesotho. In the case of Boipuso, a diesel generator is used in the village, but it supplies electricity to only half of the households, with the rest using wood as main energy source. From the literature review, it was found that extending the grid to those regions would not be economically feasible due to their remote locations.

After the selection of the villages, modelling of the HRES in HOMER required the evaluation of the solar and wind resources on each site. This was carried out through a resource assessment plan. The global horizontal solar irradiance (GHI) data and the monthly average wind speed data at an anemometer height of 50m for each village were obtained from NASA's Surface meteorology database and were used as inputs to HOMER. This is presented in the following sections.

A. Ncera Village – Resource Assessment

Located at 33°7'4.1556" South, 7°34'5.4264" East in the Eastern Cape, Ncera is favored by strong winds with an annual average wind speed of 7.1 m/s and has an annual average solar irradiation estimated at 4.82 kWh/m²/day with sun availability of over 300 days a year. Fig. 1 and Fig. 2 show the monthly average solar radiation and monthly average wind speed at the village respectively. It is observed that in winter (April-

September), the wind resources would complement the low solar radiation levels.

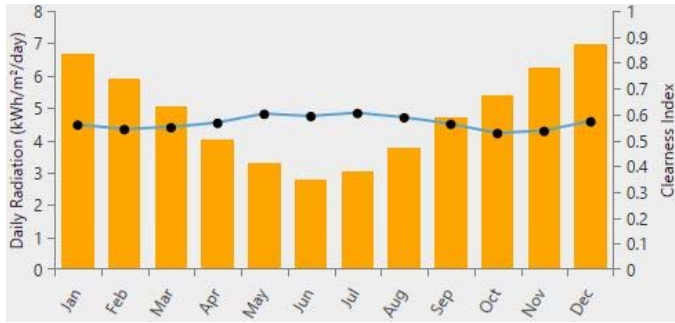


Fig. 1. Monthly average solar radiation at Ncera (kWh/m²)



Fig. 2. Monthly average wind speed at Ncera (m/s)

B. Boipuso Village – Resource Assessment

In the case of Boipuso, the village has a slightly higher annual average solar radiation at 5.52 kWh/m²/day but with a smaller annual average wind speed of 5 m/s. This is represented by Fig. 3 and Fig. 4 respectively.



Fig. 3. Monthly average solar radiation at Boipuso (kWh/m²)



Fig. 4. Monthly average wind speed at Boipuso (m/s)

III. LOAD MODELLING OF THE VILLAGES

Developing a realistic load profile is important in the accurate sizing of the system components of the HRES model.

However, due to the absence of real load data on each site, the residential load consumption for a rural household, as indicated in Table I, was assumed based on various previous studies [16] [17] [18] [19] while taking into consideration the basic needs of a family in terms of cooking, lighting, heating and entertainment. Table II shows the daily hourly load demand of the rural household model during summer and winter seasons. The hourly load varies in summer and winter due to the seasonal uses of the heater and fan.

TABLE I. ESTIMATED LOAD CONSUMPTION OF A TYPICAL RURAL HOUSEHOLD

Appliance	Power (kW)	Usage Time (hr)	Quantity	Energy (kWh/day)
Refrigerator	0.15	24	1	3.6
Radio	0.015	2	2	0.06
Electric Stove	1	1.5	1	1.5
Kettle	1	1.5	1	1.5
light bulbs	0.018	4	6	0.432
Television	0.06	8	1	0.48
Electric Iron	1	2	1	2
Heater*	1	3	1	3
Fan*	0.075	4	1	0.3
TOTAL				12.872

TABLE II. HOURLY LOAD CONSUMPTION OF RURAL HOUSEHOLD

Time of Day	Load(kWh)-Summer	Load(kWh)-Winter	Time of Day	Load(kWh)-Summer	Load(kWh)-Winter
00.00-01.00	0.06	0.1	12.00-13.00	0.15	0.15
01.00-02.00	0.06	0.1	13.00-14.00	0.15	0.15
02.00-03.00	0.06	0.1	14.00-15.00	0.11	0.11
03.00-04.00	0.12	0.12	15.00-16.00	0.14	0.14
04.00-05.00	0.22	0.22	16.00-17.00	0.21	0.21
05.00-06.00	0.35	0.35	17.00-18.00	0.55	0.55
06.00-07.00	0.42	0.42	18.00-19.00	0.65	0.65
07.00-08.00	0.38	0.38	19.00-20.00	0.45	0.45
08.00-09.00	0.28	0.28	20.00-21.00	0.35	0.35
09.00-10.00	0.21	0.21	21.00-22.00	0.28	0.28
10.00-11.00	0.188	0.18	22.00-23.00	0.1	0.2
11.00-12.00	0.21	0.21	23.00-24.00	0.1	0.2

A. Load profile of Ncera village

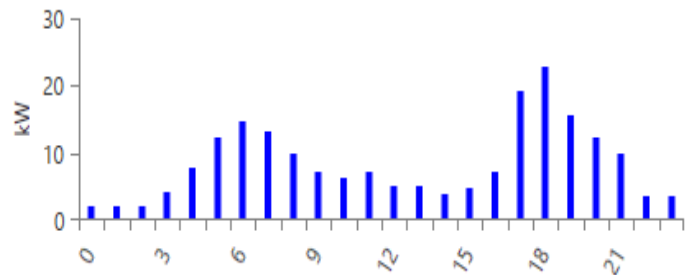


Fig. 5. Load profile for Ncera (kW/hr. of day)

Ncera consists of 30 households with a total population of 150 villagers. The village also consists of three taverns which have similar load profiles as a typical rural household. Since load growth is anticipated, two extra households were included, and the village was modelled with a total of 35 households. The load profile of the village as shown in Fig. 5 was then generated in HOMER using the values in Tables I and II.

B. Load profile of Boipuso Village

Boipuso consists around 75 houses with a population of 600 villagers. The village consists of a church, a small clinic, a public elementary school and two small stores. The load profile of the stores and the church were approximated to that of a rural household and were therefore added up to the residential load for a total of 80 households. The estimated load demand of the school was estimated at 9.32 kWh/day based on [20]. The estimated load demand of the clinic was 10.45 kWh/day referenced from [21]. The load profile of the village was then generated as shown in Fig. 6 using the load data as shown in Table III.

TABLE III. LOAD DATA OF NCERA AND BOIPUSO

Metric	Ncera	Boipuso
Population	150	600
Average demand (kWh/day)	207.34	518.39
Peak demand(kW)	39.33	95.91
Load factor	0.22	0.23

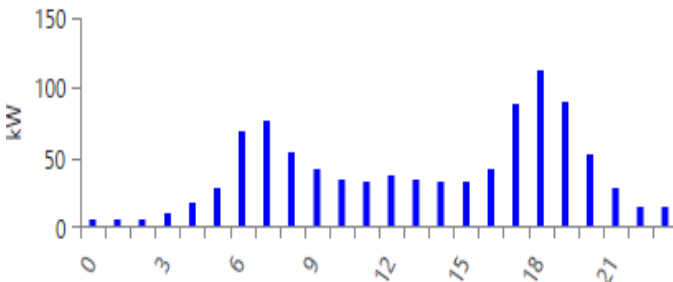


Fig. 6. Load profile for Boipuso (kW/hr. of day)

IV. HRES DESIGN

A. Selection of System Components

From the resource assessment, it was found that both selected villages were suitable for the implementation of an HRES comprising of a PV and wind turbine system. A battery system as well as a diesel generator were included in the design of the standalone energy system. The diesel generators were included to ensure uninterrupted power supply and they were selected so that their rated capacity was closely related to the peak power demand of the villages. The battery was added to store any excess energy from the power system and to be used as backup supply. Table IV provides a technical and economic summary of the system components used in the design of the HRES.

TABLE IV. SYSTEM COMPONENT SPECIFICATIONS

PV module: Canadian MAXPOWER CS6-325P	
Capital cost (\$/kW)	2500
Replacement cost (\$/kW)	2200
O&M cost (\$/kW)	0
Lifetime/years	25
Derating Factor (%)	88
Module efficiency (%)	16.94
Ground reflectance (%)	20
Wind turbine: Eocycle EO20	
Capital Cost (\$)	55000

Replacement cost (\$)	50000
O&M Cost (\$)	800/ year
Lifetime (years)	20
Hub height (m)	36
Rated Capacity (kW)	20
Battery: Trojan IND29-4V	
Capital Cost (\$)	1171.09
Replacement cost (\$)	1000
O&M cost (\$/year)	11.7
Lifetime /years	17
Roundtrip efficiency (%)	80
Converter	
Capital cost (\$)	300
Replacement cost (\$)	300
O&M cost (\$)	0
Diesel Generator: CUMMINS 4BT3.9G2(Ncera)	
Capital cost (\$)	7917.8
Replacement cost (\$)	7917.8
O&M cost(\$/hr.)	2
Lifetime (hours)	15000
Rated capacity (kW)	44
Diesel Generator: PERKINS GP-P110-60 (Boipuso)	
Capital cost (\$)	25980
Replacement cost (\$)	20000
O&M cost(\$/hr)	2
Lifetime (hours)	15000
Rated capacity (kW)	100

B. HOMER Inputs and Optimization Criteria

The data from the resource assessment shown in Fig. 1, Fig. 2, Fig. 3, Fig. 4 and load data of the villages as depicted in Table III were used as inputs to HOMER. The details of the selected system components shown in Table IV were also used as inputs to the software prior to simulation. HOMER can then simulate the long-term operation of an off-grid hybrid system consisting of a combination of PV modules, wind turbines, storage systems, diesel generators and storage devices. Optimization occurs automatically during the simulation process by sizing of the above-mentioned systems accurately. The economic parameters used for the optimization process include the following:

- Net Present Cost (NPC): It refers to the difference between the present value of the revenue and costs of the system over its lifetime. HOMER uses this parameter as its main economic output and ranks all the feasible system configuration in order of increasing NPC.
- Cost of Energy (COE): It refers to the average cost/kWh of useful electrical energy produced by the system.

The optimization process also considers the reliability of the system towards the load. The system must be able to meet the load requirements throughout the year. Reliability is assessed with the following parameters:

- **Unmet load:** This refers to the electrical load that the system is not able to serve and occurs when the demand exceeds the supply.
- **Capacity Shortage:** A capacity shortage occurs when the operating capacity of the system is lower than the required operating capacity.
- **Excess Electricity:** This refers to the surplus electrical energy generated by the hybrid system, but which cannot be used to serve the electric load. It is indicative of the system efficiency. An excess electricity of less than 50 % was deemed acceptable in this study to account for load growth.

The results of the simulations are ranked in order of increasing net present cost and the infeasible ones are discarded by HOMER. The results also present values on the above reliability indices and allows the designer to select the optimal system configuration.

V. CASE STUDIES

In this work, six case studies were simulated for each village in HOMER as listed in Table V. They were assessed based on the economic and reliability indices described in section IV.

TABLE V. CASE STUDY DESCRIPTION

Case Study	Configuration
I	Standalone DG
II	Solar PV/DG/Battery
III	Wind turbine/DG/Battery
IV	Solar PV/Wind Turbine/Battery
V	Solar PV/Wind Turbine/DG
VI	Solar PV/Wind Turbine/DG/Battery

Case studies I-VI were modelled for both villages. Case study I was selected as the base case since this configuration is widely used in rural settlements across South Africa. Case studies II and III consisted of a solar PV or wind turbine respectively, hybridized with a diesel generator and battery. These configurations were used to assess whether a single renewable energy source could meet the load requirements of the villages. Finally, case studies IV-VI were composed of two renewable energy sources hybridized with both a diesel generator and battery system.

VI. RESULTS AND DISCUSSION

The simulations were carried out with discount rate of 8% and diesel price of \$1.06/litre.

A. Results for Optimal HRES for Ncera Village

The key technical and economic parameters for the optimal HRES obtained from HOMER simulation for Ncera are summarized in Table VI.

TABLE VI. TECHNICAL AND ECONOMIC RESULTS FOR NCERA

Case Study	Total NPC (\$)	COE (\$/kWh)	Excess energy (kWh)	Unmet load (kWh)	Capacity shortage (kWh)	CO2 emission (kg/yr)
I	1094398	0.911	34082	0	0	110589
II	341318.7	0.284	14803	0	0	3895
III	215786	0.18	10913	0	0	3280
IV	233450	0.1944	58113	36.7	62.7	0
V	629317	0.524	265356	0	0	33306
VI	208308.8	0.173	23121	0	0	1571

From an economic perspective, the base case is not feasible with its high NPC and COE values and this is attributed to the high cost of fuel required to run the system. The most economic optimal configuration for the village of Ncera is case study VI, shown in Fig. 7. The system consists of a 7.80 kW rated PV system, a 20-kW rated wind turbine, a 50-kW convertor, a 44-kW diesel generator and 45 units of Trojan IDN29 batteries. It has the lowest NPC and COE of \$208308.8 and \$0.173/kWh respectively which is about five times lower than that of the base case. This makes use of maximum hybridization of the locally available renewable along with diesel generator and battery banks.

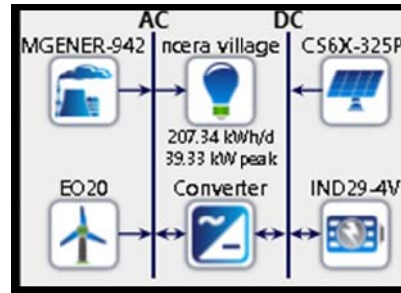


Fig. 7. Solar PV/Wind turbine/DG/Battery system for Ncera

Case study IV also produces the second lowest excess electricity at 21.4% of the total electrical production of the system after case study III which achieved the lowest value but with a higher NPC and COE. Even though the system is slightly oversized, the excess electricity is within acceptable limits and can be used for future load growth. Fig. 8 shows that the optimal configuration can provide a steady supply of power throughout the year with a wind turbine share of 86.8%, PV share of 11.8% and a share of 1.42% from the diesel generator. The limited usage of the DG allowed for the lowest CO2 emissions when compared to the other case studies which involved the DG.

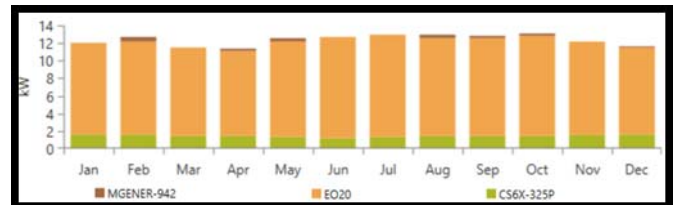


Fig. 8. Monthly average electric production of PV/WT/DG/battery system for Ncera

B. Results for Optimal HRES for Boipuso Village

TABLE VII. TECHNICAL AND ECONOMIC RESULTS FOR BOIPUSO

Case Study	Total NPC (\$)	COE (\$/kWh)	Excess energy (kWh)	Unmet load (kWh)	Capacity shortage (kWh)	CO2 emission (kg/yr)
I	1984974.8	0.661	67403	0	14.6	233960
II	726037.9	0.242	21027	0	0	9702
III	737462.2	0.246	85003	0	0	9282
IV	657122.2	0.218	49451	112	181	0
V	1563753.5	0.520	294042	0	0	126438
VI	630009.4	0.210	42405	0	0	6001

The key technical and economic parameters for the optimal HRES calculated using HOMER simulation for Boipuso are summarized in Table VII. Similar to Ncera, case study VI as shown in Fig. 9 is the optimal configuration for Boipuso. The system consists of a 70.3 kW rated PV system, two 20-kW rated wind turbines, a 75-kW convertor, a 100-kW diesel generator and 102 units of Trojan IDN29 batteries. The total NPC and COE of the system is \$ 630009.4 and \$0.210/kWh respectively which is about 3 times lower than that of the base case. It is the cheapest configuration which can meet the load demand of the village effectively with zero capacity shortage and unmet load. The system also has the second lowest excess electricity which amounts to 16.4% of the total electrical production. Case study II achieved the lowest excess electricity but has a higher NPC and COE. When comparing case studies V and VI, it is observed that adding a storage scheme to the system not only allowed for a reduction in excess electricity of the system but also in its NPC and COE.

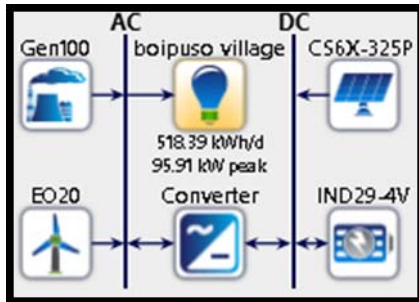


Fig. 9. Solar PV/Wind turbine/DG/Battery system for Boipuso

The monthly electricity production of the optimal system is shown in Fig. 10. 51.9% of the total electrical production is provided by the PV system while the wind turbine system contributes with 45.6%. The remaining 2.45% of the electrical production is provided by the diesel generator. Similar to the optimal case for Ncera, the optimal configuration for Boipuso produced the lowest CO2 emissions when compared to the other cases which included a DG.



Fig. 10. Monthly average electric production of PV/WT/DG/battery system for Boipuso

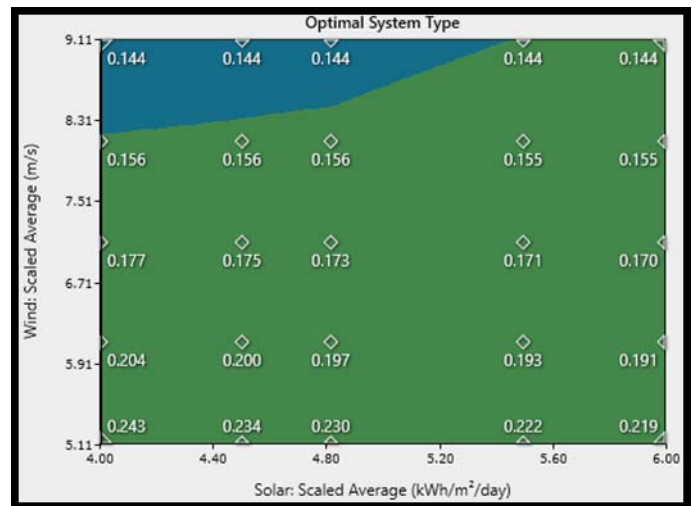
VII. SENSITIVITY ANALYSIS

A sensitivity analysis is also carried out here using two sensitivity parameters namely, annual average wind speed and the annual average solar radiation with values listed in Table VIII. The effects of varying the sensitivity parameters were analyzed using the optimal system type graphs with the COE superimposed on the graph. Fig. 11 (a) and (b) illustrates the results for Ncera and Boipuso respectively.

TABLE VIII. Sensitivity Parameters for Ncera and Boipuso

Solar Scaled Average (kWh/m ² /day)		Wind Scaled Average (m/s)	
Ncera	Boipuso	Ncera	Boipuso
4.816	5.51	7.11	5.00
4.334	4.96	6.11	4.00
3.900	4.46	5.11	3.00
3.510	4.02	4.11	2.00

The results showed in green in Figure 11 indicate that the optimal HRES for both villages still is the Solar PV/Wind turbine/DG/Battery system for most of the variations in the annual average wind speed and annual average solar radiation.



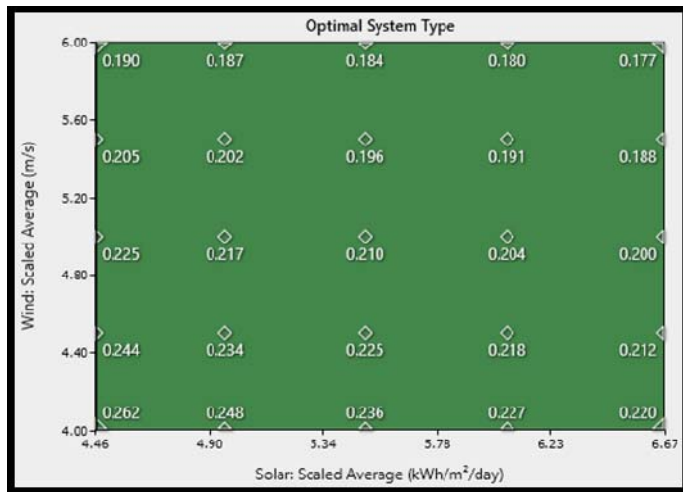


Fig. 11. Optimal HRES (a) Ncera (b) Boipuso after sensitivity analysis

VIII. CONCLUSION

From the optimization results, it is noted that a combination of diesel generator and battery with the renewable energy source reduces the NPC and COE of the system. Hybridizing a second renewable energy source to the system allowed for a further reduction in the economic parameters. Therefore, based on these results, a Solar PV/wind turbine system hybridized with a diesel generator and a battery storage system was found to be the most cost-effective solution in providing a reliable source of electricity to the selected villages. The effects of varying annual wind speed and annual average solar radiation were investigated through a sensitivity analysis. The sensitivity results agreed with the optimization results and indicated that the Solar PV/Wind/DG/battery systems were the optimal ones with variations in the solar and wind resources. Access to electricity would help to boost the productivity of the villagers to promote future village developments which would support poverty alleviation.

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